Answers: Introduction to logic (maths)

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Summary

Answers to questions relating to the guide on introduction to logic (maths).

These are the answers to Questions: Introduction to logic (maths).

Please attempt the questions before reading these answers!

Q1 Propositions

1.1. "$3 < 5$" is a proposition.

1.2. "$5 < 3$" is a proposition.

1.3. "$x < 5$" is not a proposition.

1.4. "5 is even" is a proposition.

1.5. "y is odd" is not a proposition.

1.6. "$2 \cdot 3 = 6$" is a proposition.

1.7. "$4 \cdot 3 = 10$" is a proposition.

1.8. "$2 \cdot 3 < 5x$" is not a proposition.

Q2 Connectives

2.1. The logical structure of "If $x < 3$, then y is even" is $P \to Q$.

2.2. The logical structure of "x is odd and positive" is $P \wedge Q$.

2.3. The logical structure of "z is not -1" is $\neg P$.

2.4. The logical structure of "y is even or odd" is $P \vee Q$.

2.5. The logical structure of "z is positive if and only if y is negative" is $P \leftrightarrow Q$.

2.6. The logical structure of "If $x < 0$, then y and z are even" is $P \to (Q \wedge R)$.

Q3 Truth tables

3.1. $P \to Q$

P	Q	$P \to Q$
T	T	T
T	F	F
F	T	T
F	F	T

3.2. $\neg P$

P	$\neg P$
T	F
F	T

3.3. $(P \lor Q) \land \neg Q$

P	Q	$(P \lor Q) \land \neg Q$
T	T	F
T	F	T
F	T	F
F	F	F

3.4. $\neg(P \leftrightarrow Q)$

P	Q	$\neg(P \leftrightarrow Q)$
T	T	F
T	F	T
F	T	T
F	F	F

3.5. $\neg(P \land Q)$

P	Q	$\neg(P \land Q)$
T	T	F

P	Q	$\neg(P \wedge Q)$
T	F	T
F	T	T
F	F	T

3.6. $(P \to Q) \to R$

P	Q	R	$(P \to Q) \to R$
T	T	T	T
T	F	T	T
T	T	F	F
T	F	F	T
F	T	T	T
F	F	T	T
F	T	F	F
F	F	F	F

3.7. $\neg(P \vee (Q \vee R))$

P	Q	R	$\neg(P \vee (Q \vee R))$
T	T	T	F
T	F	T	F
T	T	F	F
T	F	F	F
F	T	T	F
F	F	T	F
F	T	F	F
F	F	F	T

3.8. $(P \leftrightarrow Q) \vee (\neg R)$

P	Q	R	$(P \leftrightarrow Q) \vee (\neg R)$
T	T	T	T
T	F	T	F
T	T	F	T
T	F	F	T
F	T	T	F
F	F	T	T
F	T	F	T
F	F	F	T

Q4 Equivalence

4.1. $\neg(P \wedge Q)$ and $(\neg P) \vee (\neg Q)$ are logically equivalent. This identity is known as **De Morgan's Law**.

4.2. $P \to Q$ and $Q \to P$ are not logically equivalent.

4.3. $P \to Q$ and $(\neg P) \vee Q$ are logically equivalent.

4.4. $P \to Q$ and $\neg(P \vee Q)$ are not logically equivalent.

Version history and licensing

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